

In[150]:=

```
Clear[LaneEmdenSolve]

R = 6.96 * 108;
LaneEmdenSolve[n_, ξmax_ : R, rhoC_ : 1.53 * 105,
  K_ : 5.556 * 109, G_ : 6.67 * 10-11, C_ : 3 * 108] :=
Module[
  {eps = 10-7,
   sol,
   a,
   theta, deltheta,
   r, rmax,
   rho, delrho,
   P, delP,
   m,
   g,
   rat, arat,
   B,
   T, delT,
   σ, ϕ, L,
   kb, mh, mu
  },

  sol =
    NDSolve[{
      θ''[ξ] + (2 / ξ) θ'[ξ] + θ[ξ]^n == 0,
      θ[eps] == 1 - eps2 / 6,
      θ'[eps] == -eps / 3
    }, θ, {ξ, eps, ξmax}] (*,MaxStepSize→0.05][[1]]*);

  a = If[
    n == 0,  $\sqrt{\frac{K}{4 \pi G \rho C}}$  (*constant-density sphere*),
     $\sqrt{\frac{K (n + 1) \rho C^{\left(\frac{1}{n} - 1\right)}}{4 \pi G}}$ 
  ];

  (*theta=θ/. sol;*)
  theta = θ /. sol[[1]];
  (*theta[ξ_]:=Evaluate[θ[ξ]/. sol];*)
  deltheta[ξ_] := theta'[ξ];

  r[ξ_] := a ξ;
  rmax = a ξmax;
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(* define xi = r/a *)
rho[r_] := rhoC theta[r/a]^n;
delrho[r_] := (rhoC n/a) theta[r/a]^(n-1) × deltheta[r/a];

(*Solve mass equation*)
m =
NDSolveValue[{
  M'[r] == 4 Pi a^2 (r/a)^2 rho[r],
  M[0] == 4 Pi rhoC (a*eps)^3/3
}, M, {r, 0, rmax}];

(*Hydrostatic equilibrium*)
P[r_] := If[
  n == 0, K*rho[r],
  K*rho[r]^(1+1/n)
];
delP[r_] := P'[r];

(*g(r)*)
g[r_] :=  $\frac{G m[r]}{r^2}$ ;

(* ratio functions *)
rat[r_] :=  $\frac{\text{delP}[r]}{\text{rho}[r] \times g[r]}$ ;
B[r_] :=  $\frac{\text{delP}[r]}{\text{rho}[r] \times g[r]} + 1$ ;

(*T curve and Eddington Emden Chandrasekhar Equation *)
kb = 1.38 * 10-23;
mh = 1.67 * 10-27;
mu = 0.61;
T[r_] :=  $\frac{\mu mh}{kb} \frac{P[r]}{\text{rho}[r]}$ ;
delT[r_] := T'[r];
(*stefan bolzmann constant σ *)
σ = 5.67 * 10-8;
(*some constant C *)
(*Flux*)
Φ[r_] :=  $-\frac{\text{delT}[r] 4 \sigma (T[r])^3}{C \text{rho}[r]}$ ;
(*Luminosity*)
L[r_] :=  $\frac{\Phi[r]}{4 \text{Pi } r^2}$ ;

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<|
  "n" → n, "Solution" → sol, "Theta" → theta, "a" → a,
  "r" → r, "rmax" → rmax,
  "Rho" → rho, "Mass" → m,
  "P" → P, "dP" → delP,
  "g" → g,
  "r1" → rat,
  "B" → B,
  "T" → T,
  "F" →  $\Phi$ , (*Flux*)
  "L" → L
|>

```

In[155]:=

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 $\xi_{\max} = 100$ ;
lane2 = LaneEmdenSolve[3.35,  $\xi_{\max}$ ];

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(*a*)
N[lane2["a"]];

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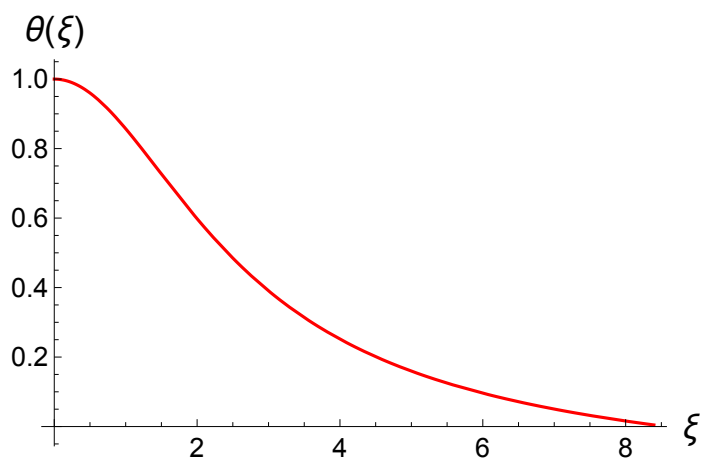
In[130]:=

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(* $\theta(\xi)$ *)
Plot[
  lane2["Theta"] [ $\xi$ ],
  { $\xi$ , 0,  $\xi_{\max}$ },
  PlotRange → All,
  PlotStyle → Directive[Red, Thickness[0.005]],
  AxesLabel → {Style[" $\xi$ ", Black, 18], Style[" $\theta(\xi)$ ", Black, 18]},
  TicksStyle → Directive[Black, 14]
]

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Out[130]=



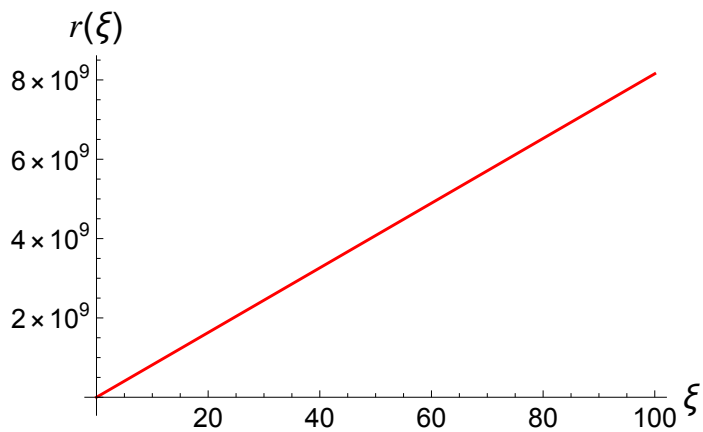
In[111]:=

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(*r(ξ)*)
Plot[
  lane2["r"][ξ],
  {ξ, 0, ξmax},
  PlotRange → All,
  PlotStyle → Directive[Red, Thickness[0.005]],
  AxesLabel → {Style["ξ", Black, 18], Style["r(ξ)", Black, 18]},
  TicksStyle → Directive[Black, 14]
]

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Out[111]=



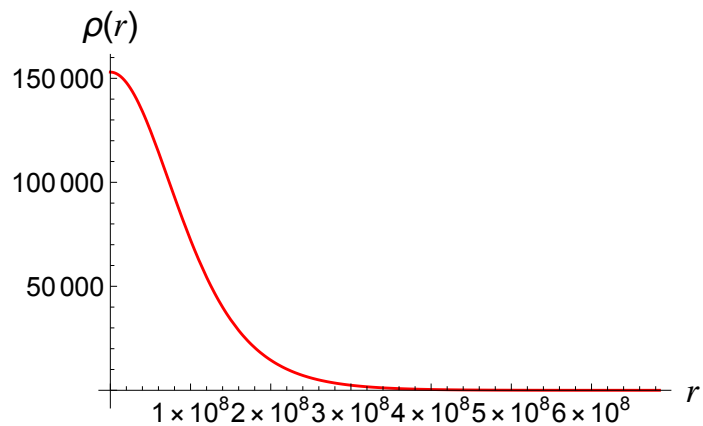
In[131]:=

```

(*ρ(r)*)
Plot[
  lane2["Rho"][r],
  {r, 0, lane2["rmax"]},
  PlotRange → All,
  PlotStyle → Directive[Red, Thickness[0.005]],
  AxesLabel → {Style["r", Black, 18], Style["ρ(r)", Black, 18]},
  TicksStyle → Directive[Black, 14]
]

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Out[131]=



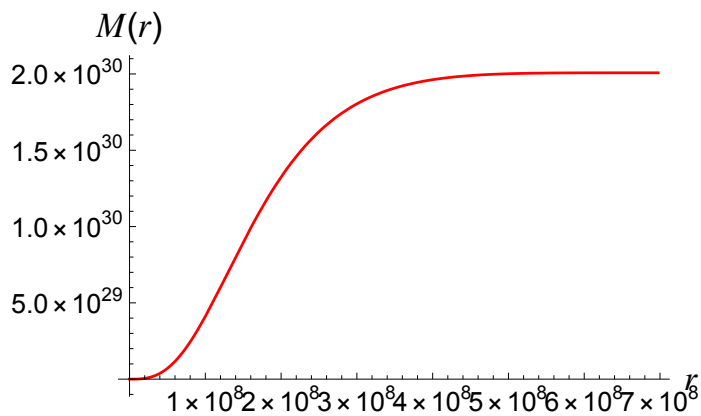
In[132]:=

```

(*M(r)*)
Plot[
  lane2["Mass"][r],
  {r, 0, lane2["rmax"]},
  PlotRange -> All,
  PlotStyle -> Directive[Red, Thickness[0.005]],
  AxesLabel -> {Style["r", Black, 18], Style["M(r)", Black, 18]},
  TicksStyle -> Directive[Black, 14]
]

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Out[132]=



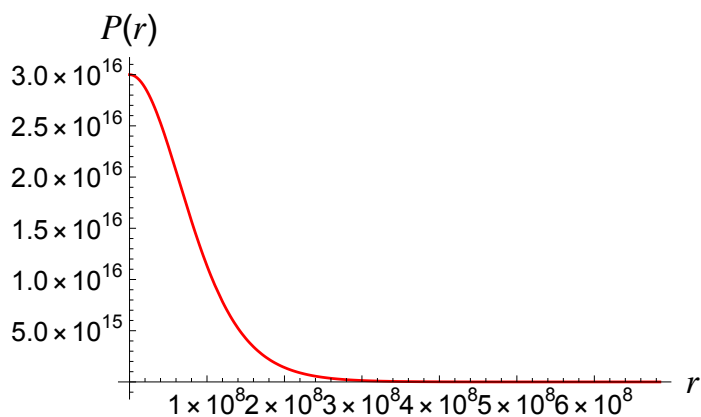
In[134]:=

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(*P(r)*)
Plot[
  lane2["P"][r],
  {r, 0, lane2["rmax"]},
  PlotRange -> All,
  PlotStyle -> Directive[Red, Thickness[0.005]],
  AxesLabel -> {Style["r", Black, 18], Style["P(r)", Black, 18]},
  TicksStyle -> Directive[Black, 14]
]

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Out[134]=



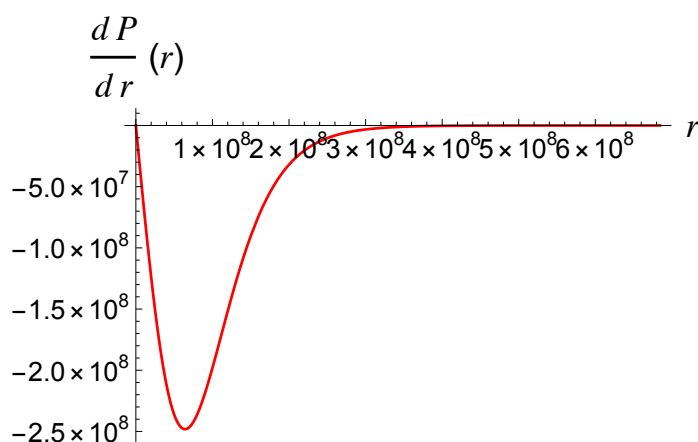
In[135]:=

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(*  $\frac{dP}{dr}(r)$  *)
Plot[
  lane2["dP"][r],
  {r, 0, lane2["rmax"]},
  PlotRange → All,
  PlotStyle → Directive[Red, Thickness[0.005]],
  AxesLabel → {Style["r", Black, 18], Style[" $\frac{dP}{dr}(r)$ ", Black, 18]},
  TicksStyle → Directive[Black, 14]
]

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Out[135]=



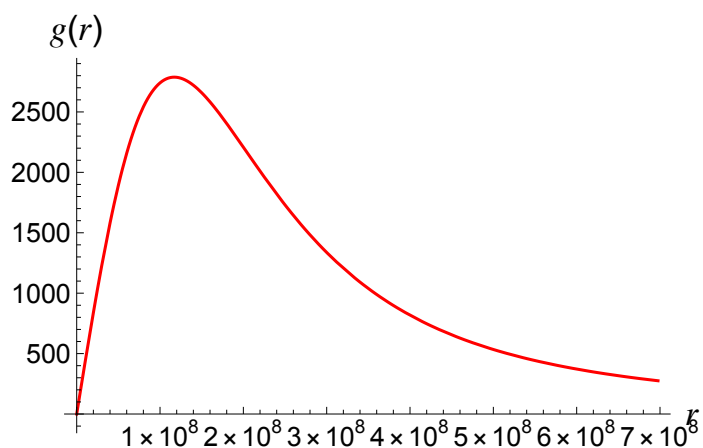
In[136]:=

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Plot[
  lane2["g"][r],
  {r, 0, lane2["rmax"]},
  PlotRange → All,
  PlotStyle → Directive[Red, Thickness[0.005]],
  AxesLabel → {Style["r", Black, 18], Style["g(r)", Black, 18]},
  TicksStyle → Directive[Black, 14]
]

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Out[136]=

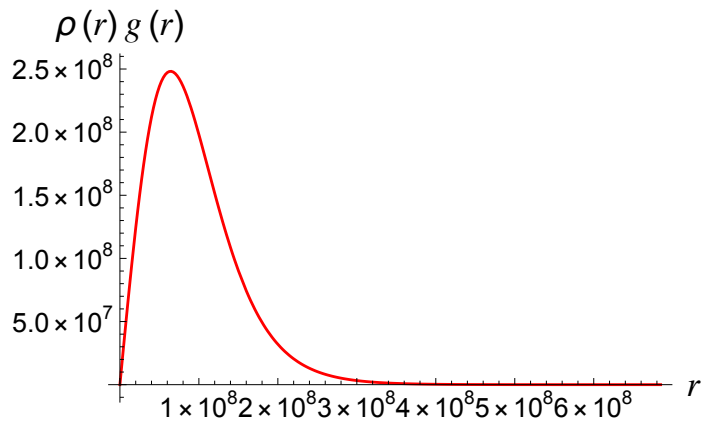


```

(* $\rho(r)g(r)$ *)
Plot[
  lane2["Rho"][r]  $\times$  lane2["g"][r],
  {r, 0, lane2["rmax"]},
  PlotRange  $\rightarrow$  All,
  PlotStyle  $\rightarrow$  Directive[Red, Thickness[0.005]],
  AxesLabel  $\rightarrow$  {Style["r", Black, 18], Style[" $\rho(r)g(r)$ ", Black, 18]},
  TicksStyle  $\rightarrow$  Directive[Black, 14]
]

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Out[137]=



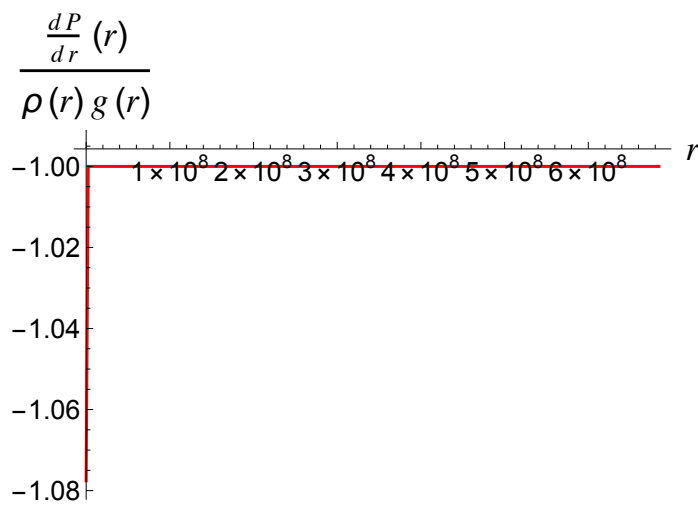
In[158]:=

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(*  $\frac{dP}{dr}(r)$  *)
Plot[
  lane2["r1"][r] ,
  {r, 0, lane2["rmax"]},
  PlotRange -> All,
  PlotStyle -> Directive[Red, Thickness[0.005]],
  AxesLabel -> {Style["r", Black, 18], Style[" $\frac{dP}{dr}(r)$ ", Black, 18]},
  TicksStyle -> Directive[Black, 14]
]

```

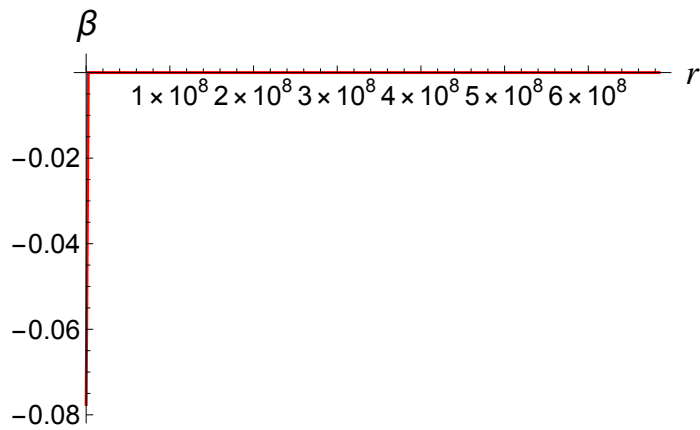
Out[158]=



In[159]:=

```
Plot[
  lane2["B"][r] ,
  {r, 0, lane2["rmax"]},
  PlotRange -> All,
  PlotStyle -> Directive[Red, Thickness[0.005]],
  AxesLabel -> {Style["r", Black, 18], Style["β", Black, 18]},
  TicksStyle -> Directive[Black, 14]
]
```

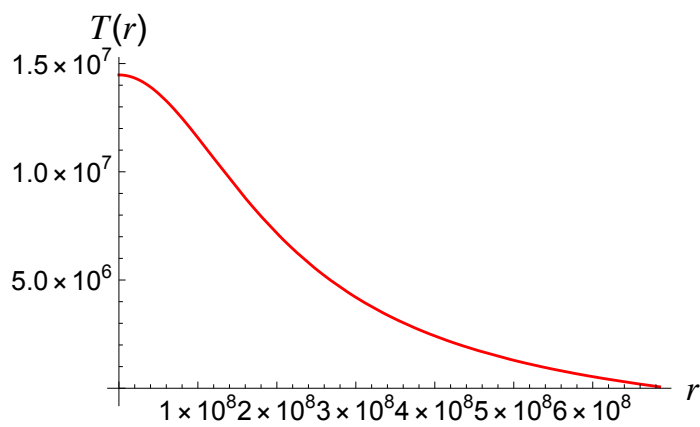
Out[159]=



In[162]:=

```
(*T(r)*)
Plot[
  lane2["T"][r] ,
  {r, 0, lane2["rmax"]},
  PlotRange -> All,
  PlotStyle -> Directive[Red, Thickness[0.005]],
  AxesLabel -> {Style["r", Black, 18], Style["T(r)", Black, 18]},
  TicksStyle -> Directive[Black, 14]
]
```

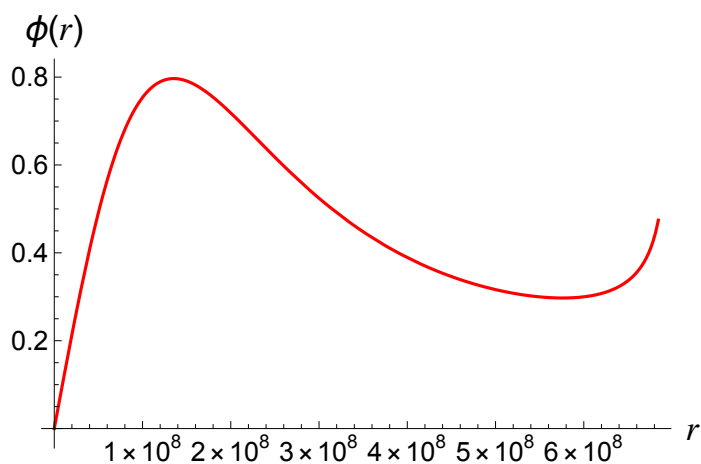
Out[162]=



In[160]:=

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Plot[
  lane2["F"][r] ,
  {r, 0, lane2["rmax"]},
  PlotRange -> All,
  PlotStyle -> Directive[Red, Thickness[0.005]],
  AxesLabel -> {Style["r", Black, 18], Style[" $\phi(r)$ ", Black, 18]},
  TicksStyle -> Directive[Black, 14]
]
```

Out[160]=



In[161]:=

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Plot[
  lane2["L"][r] ,
  {r, 1, lane2["rmax"]},
  PlotRange -> All,
  PlotStyle -> Directive[Red, Thickness[0.005]],
  AxesLabel -> {Style["r", Black, 18], Style[" $L(r)$ ", Black, 18]},
  TicksStyle -> Directive[Black, 14]
]
```

Out[161]=

